

SAT 3812: Cyber Security I

A Quick Introduction to Web
Development

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Resources for learning

- Learn HTML in 15 Minutes
 - https://www.youtube.com/watch?v=Ggh_y-33Eso
- Learn CSS in 12 Minutes
 - <https://www.youtube.com/watch?v=0afZj1G0BIE>
- Learn JavaScript in 12 Minutes
 - https://www.youtube.com/watch?v=Ukg_U3CnJWI
- Learn PHP in 15 Minutes
 - <https://www.youtube.com/watch?v=ZdP0KM49IVk>
- HTML/CSS/Javascript/PHP tutorial from w3schools
 - <http://www.w3schools.com/html>
 - <http://www.w3schools.com/css>
 - <http://www.w3schools.com/php>
 - <http://www.w3schools.com/js>

Resources: Lecture notes and Cheat Sheet

- Lecture notes and cheat sheet are available on canvas for web topics below:
 - HTML
 - CSS
 - Javascript
 - PHP

- How to use them?
 - You need to learn basic knowledge with lecture notes.
 - Cheat sheet helps you on language details.

A brief history of Web Development Technologies

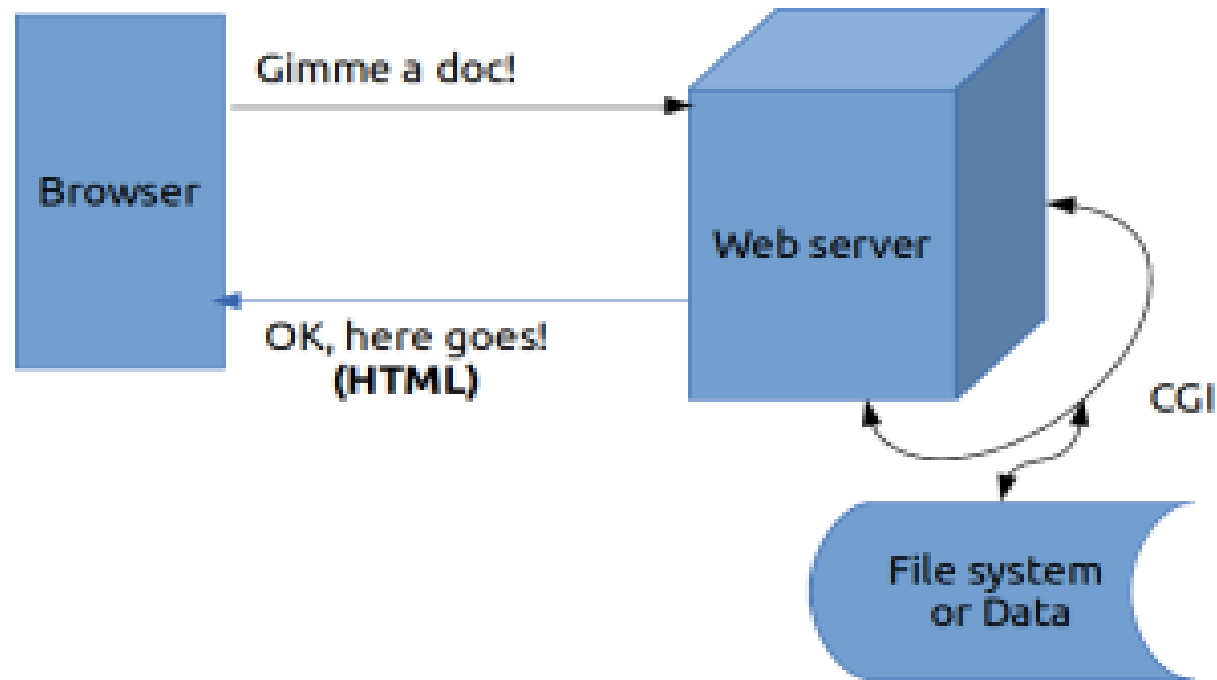
1. In the good old days: static web page.

HTML + GIF/JPG Images.



A brief history of Web Development Technologies

2. CGI(Perl): dynamic webpage, but hard to maintain, and didn't scale well.

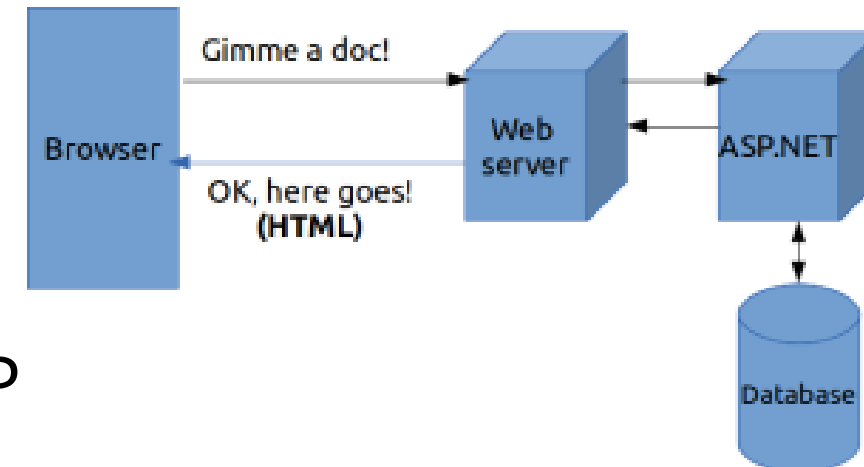


A brief history of Web Development Technologies

3. PHP / ASP / JSP: dynamic webpage. Fast development and easy to maintain. (2000 dot.com bubble)

Three popular backend choices

- LAMP: Linux+Apache+MySQL+PHP
- Microsoft: Windows+IIS+SQL Server+ASP
- Java: Linux+ Apache Tomcat +JSP+DB
- HTML + CSS + Javascript for frontend



A brief history of Web Development Technologies

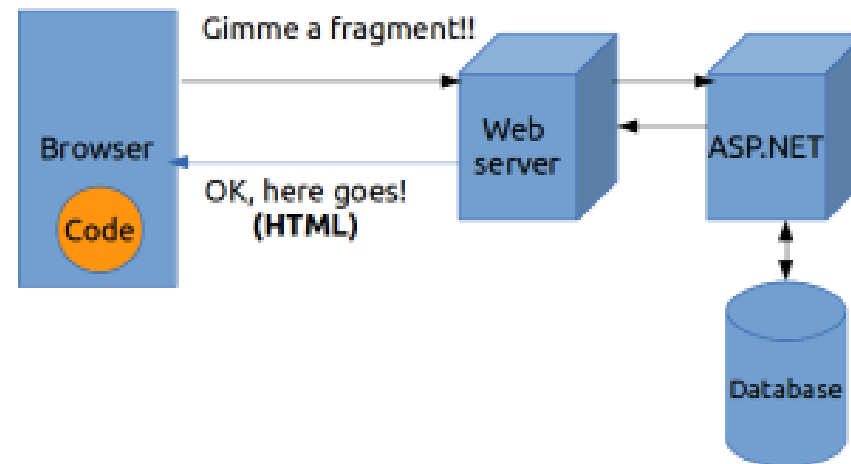
4. Web Dev after 2005.

AJAX (2005): dynamically refresh a portion of the webpage.

Like google gmail and google maps. Built based on Javascript.

⇒ Web 2.0

⇒ Web Dev starts to get complicated and professional!



A brief history of Web Development Technologies

- Smart phones and tablets:
 - Responsive design – change web page layout / content based on user devices / browsers
 - PhoneGap: a tool to wrap web application into mobile app.
- Javascript become hot:
 - JQuery: a powerful javascript library. Make it easy to build dynamic, beautiful web application.
 - AngularJS: another powerful front end javascript framework
 - Nodejs: back-end development with Javascript.
 - JSON: data interchange format

Overview of Web Dev Technologies

- Front end developer

Newbies:

- HTML + CSS + Javascript (3 good buddies)
 - More on Javascript : JQuery + Ajax
- Dead: Flash, Silverlight, Java applet...

Professionals:

- Javascript:
 - Frontend framework: AngularJS
 - Other choices: Knockout.js; Backbone.js
 - Automation tools: Grunt
- CSS
 - Frontend Framework: Bootstrap
 - Tools: SASS, Stylus
- Debug
 - Firebug; Chrome Developer Tools

Overview of Web Dev Technologies

- Back end developer
 - PHP
 - Content Management System:
 - WordPress or Drupal or Joomla
 - .NET (C#)
 - Java
 - Python
 - Framework: Django or Flask
 - Node.js
 - Javascript on backend
 - Ruby on Rails
 - Very hot with startups
 - TCL/TK
 - Popular in big companies
- Must know for backend developer
 - Database connection
 - Security
 - Performance Tuning and optimization
 - Web services / Web API / Cloud services

Overview of Web Dev Technologies

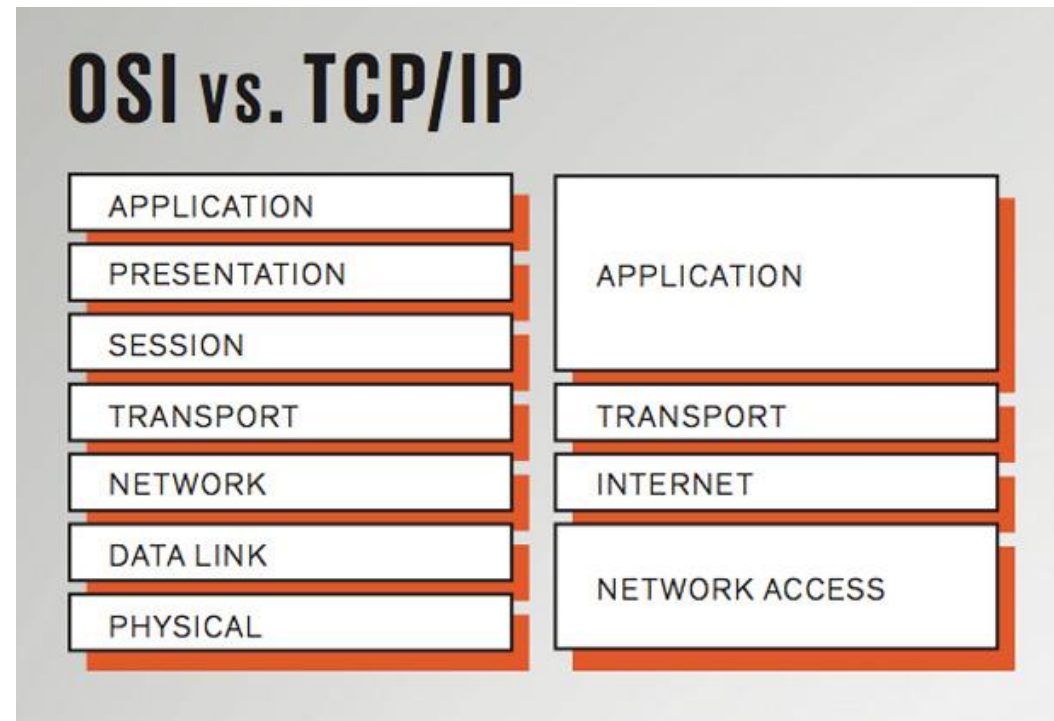
- Additional Skills for web dev
 - Database:
 - Mysql / SQL Server / Oracle
 - MongoDB (big data!)
 - Redis (tiny database)
 - Webserver:
 - Apache / IIS
 - nginx (hot!)
 - E-commerce:
 - Many choices: ZenCart, OpenCart...
 - Credit card: Authorize.Net
 - SSL and certificates
 - MVC model: model-view-controller
 - Github.com; stackoverflow.com
 - Domain name registration / web hosting
 - Photoshop and basic video editing
 - Unit test: code testing
- AI changes everything: Vibe Coding
- Job sites:
 - monster.com, dice.com, indeed.com

How much costs to build a website for a small business? (Just my personal view)

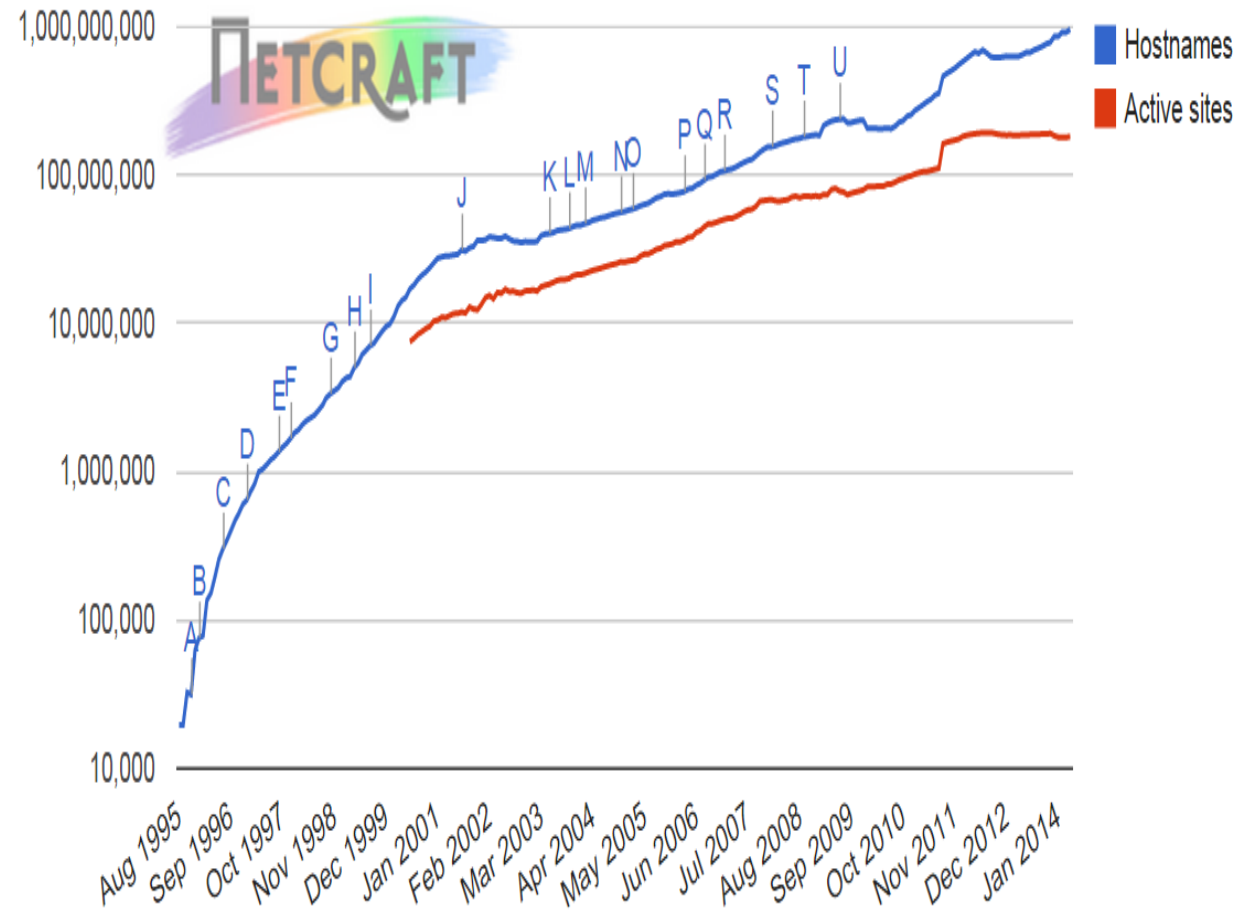
- Static website: plain HTML / Images
 - Frontend developer: average salary \$60 per hour, range \$20 - \$120.
 - Usually charge on pages; or on project
 - Usually a small business website costs \$500-\$2000.
- Dynamic website: database + dynamic content
 - Frontend developer + backend developer
 - Templates are available for small business e-commerce sites.
 - Usually more expensive to customize
- Work flow:
 - Developer collects user requirements, give 5-10 templates to pick, customers find what they like
 - Developer implement the template; work closely with graphic/art designer
 - Feedback and continuous improvement
 - Clean code!
 - You can find developers on elance.com

Yellow Tag Question

- Why TCP/IP became the standard of network?
 - Easy to implement and maintain; acceptable performance
 - Achievability on the ubiquitous platforms of the time, the atrocious combination of IBM's PC and Microsoft's DOS operating environment.
 - » TCP/IP hides the differences among devices and systems connected to the network

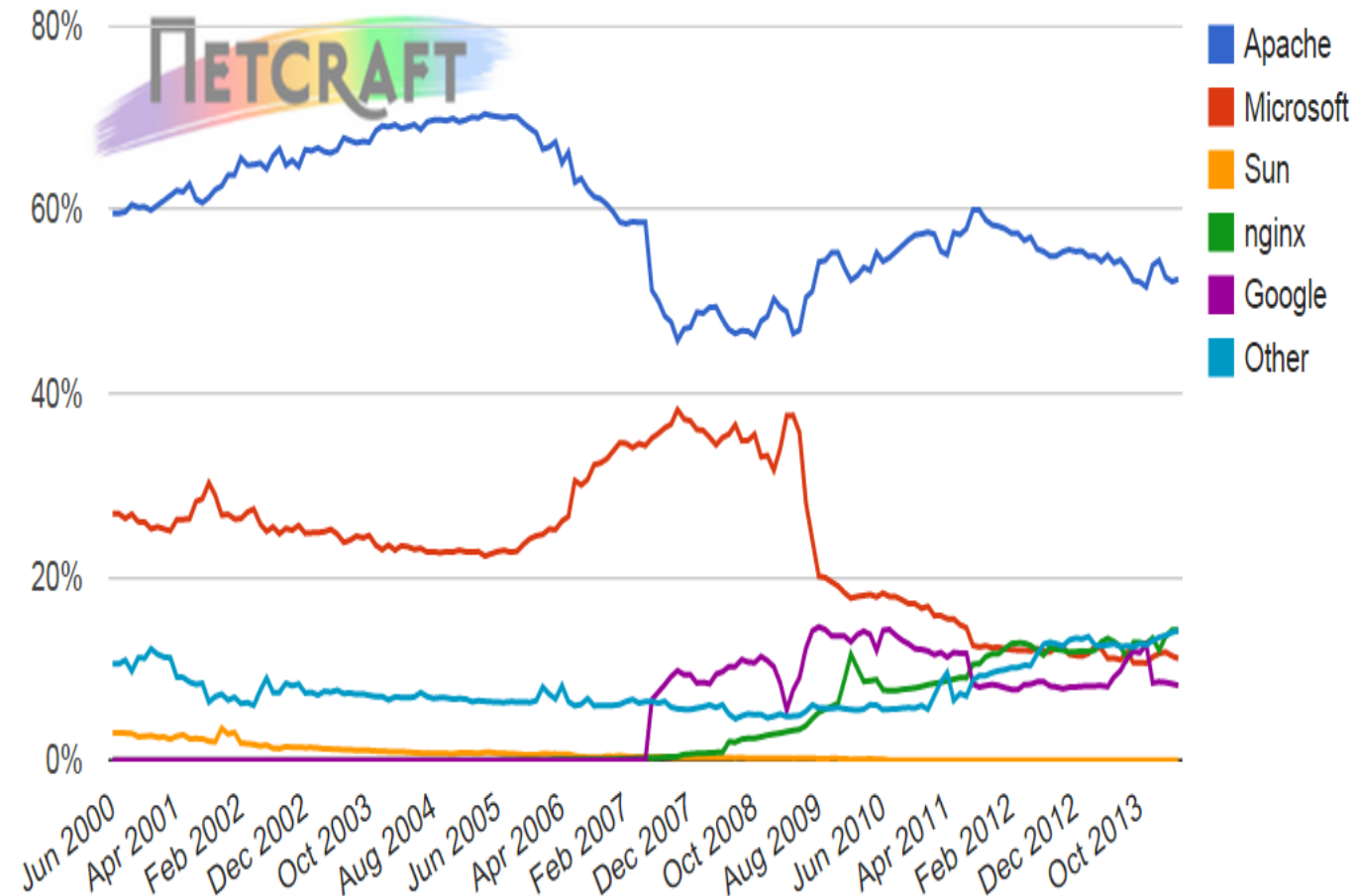


Total number of websites (logarithmic scale)



Why gap between hostnames and active sites?

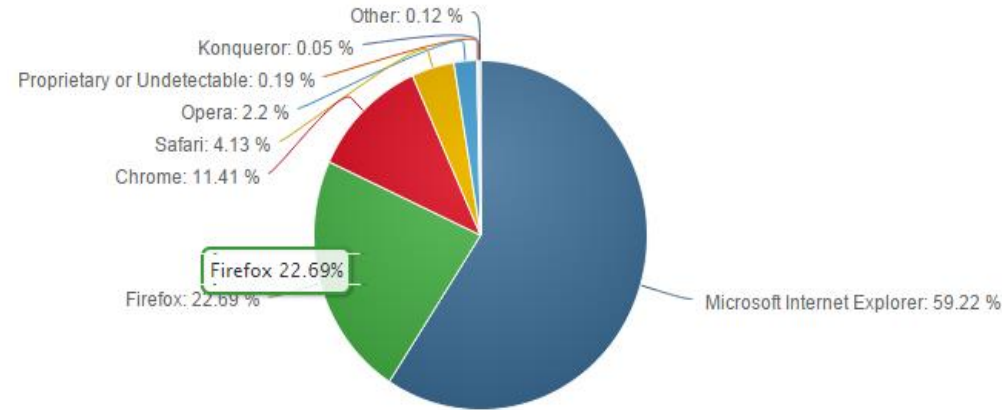
Market share of web servers among all sites



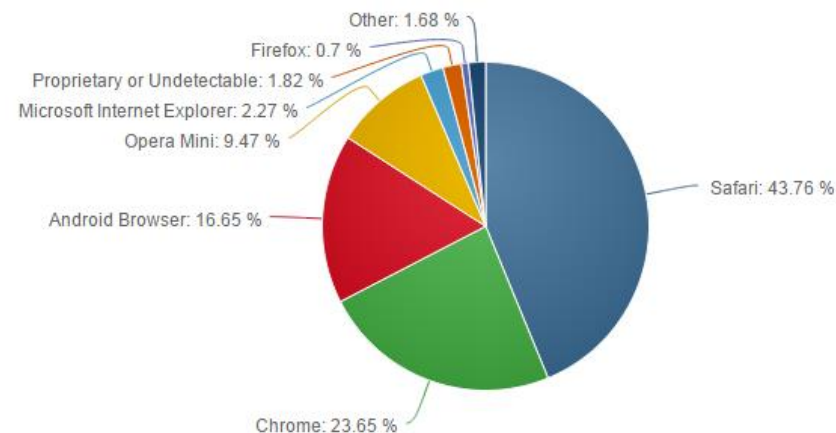
Who is the rising star?

Desktop and Mobile Brower Market Share

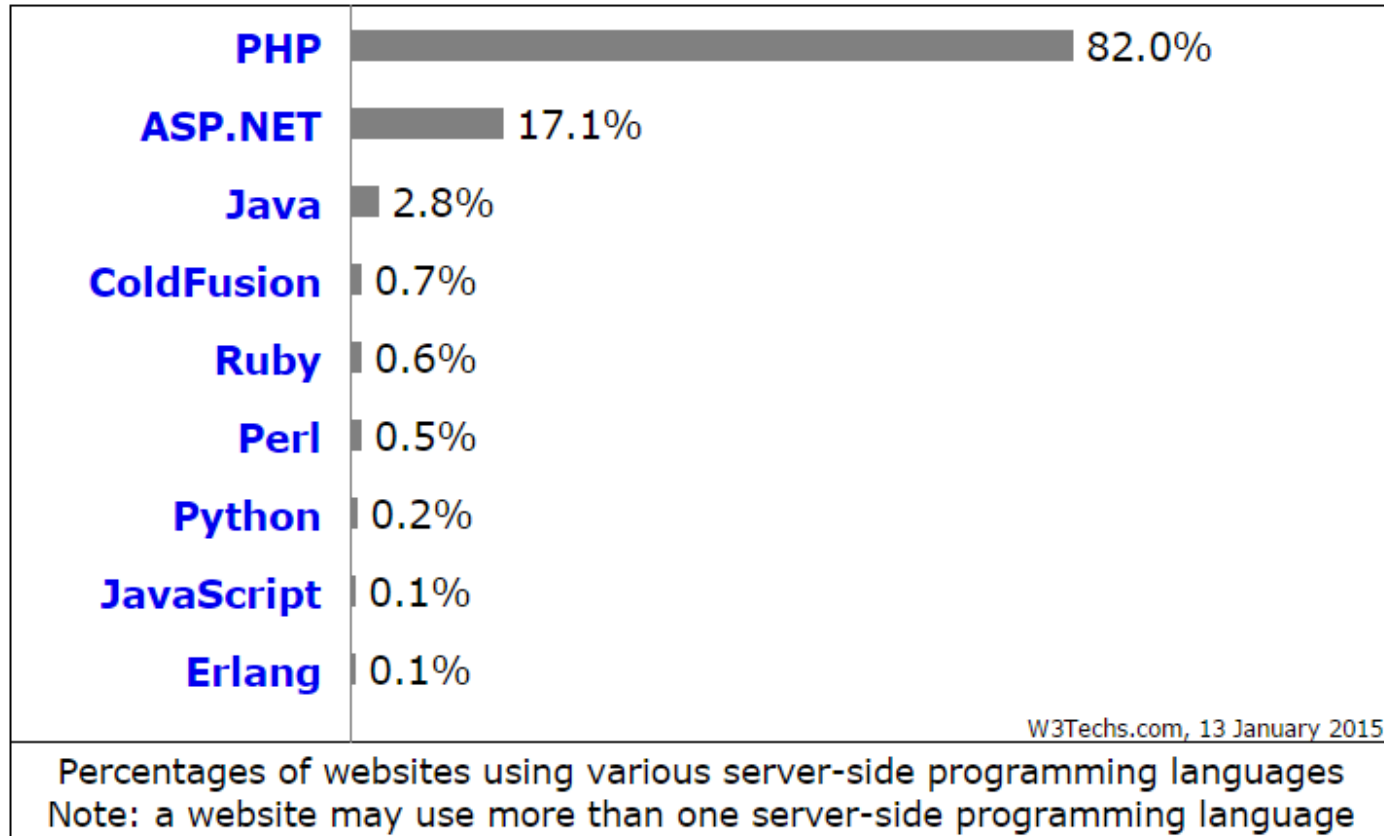
- Desktop Browser Market Share



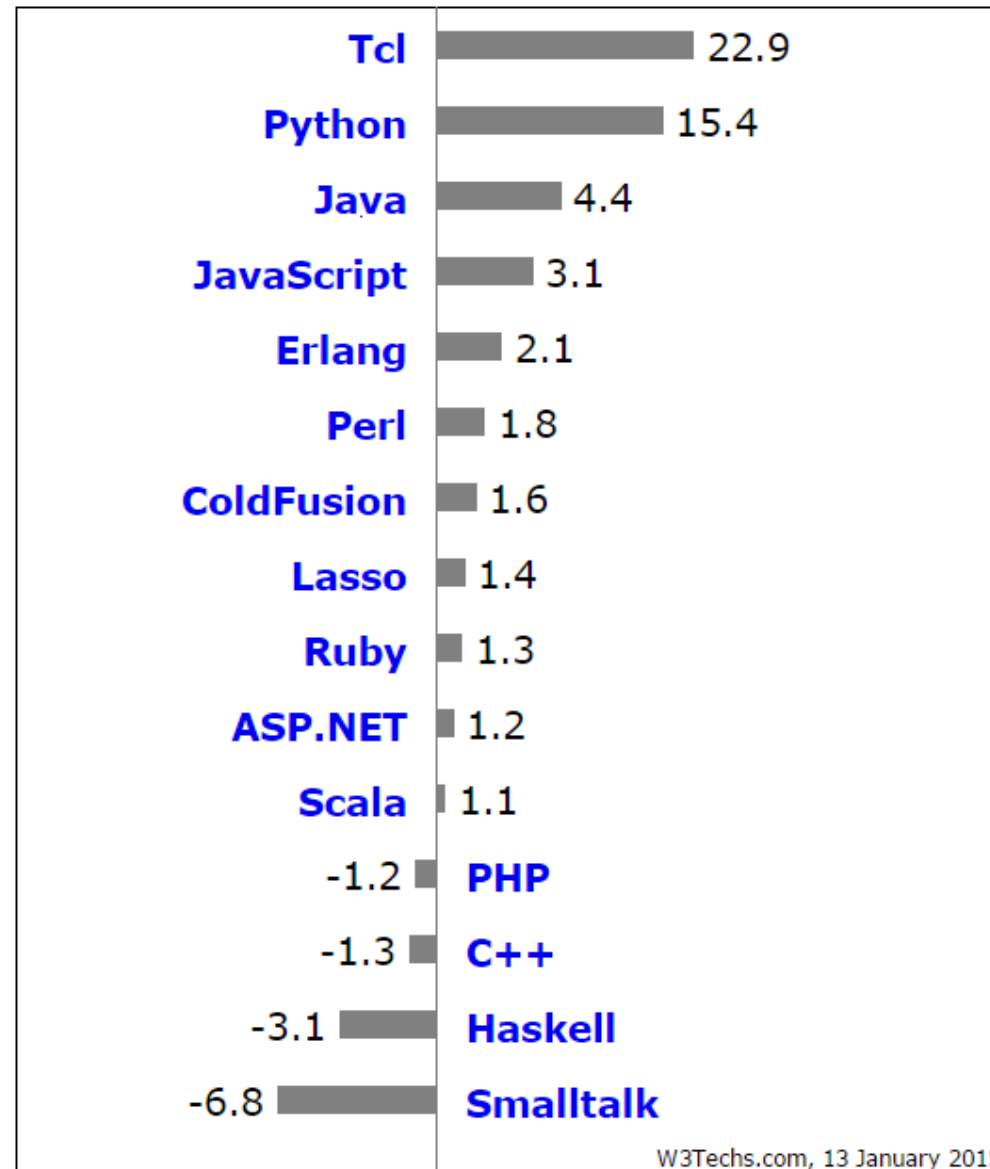
- Mobile/Tablet Browser Market Share



Usage of server-side programming languages for **all websites**

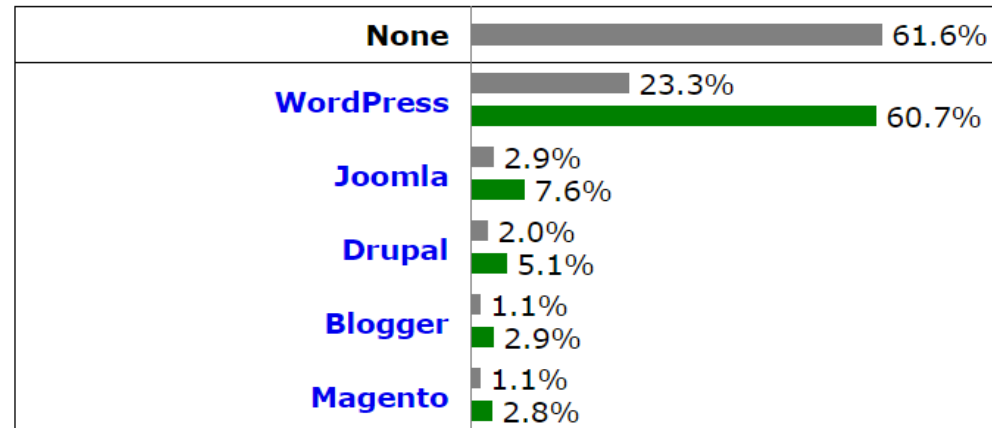


Top-site scores for server-side programming languages

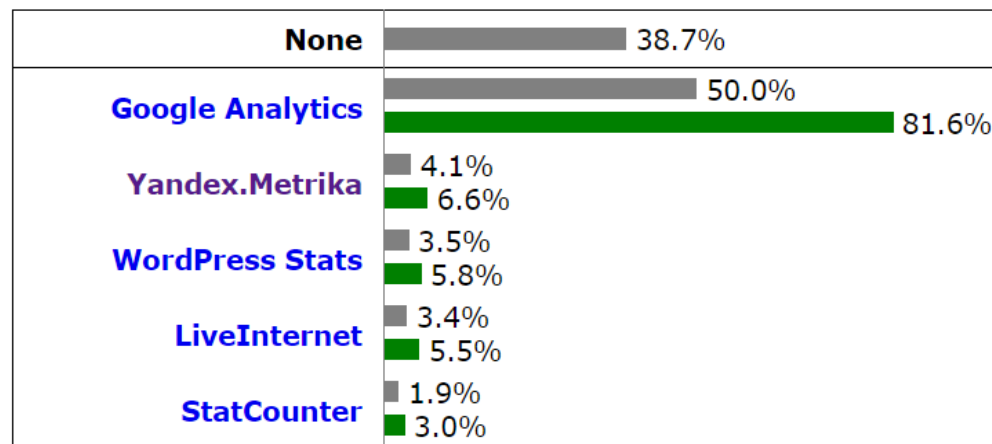


Usage of content management systems/traffic analysis tools for websites

- Usage of content management systems for websites



- Usage of traffic analysis tools for websites



Some terms

- Uniform Resource Locators (URL): a notation to specific where resource locates
- Examples:
 - `http://www.google.com:8080/some/pic.html`
- Can be in many other formats
- Multipurpose Internet Mail Extensions (MIME): tell browser the file type of web data: content-type.
 - Examples: `text/plain`, `text/html`, `image/gif`, `image/jpeg`
- Why? Because browsers can handle the file differently.
- HyperText Transfer Protocol (HTTP): protocol used by ALL Web communications
- `http://` and `https://`

The HyperText Transfer Protocol (continued)

- An example of a complete response header

Request
headers

```
Connection: Keep-Alive
User-Agent: Mozilla/4.74 [en] (WinNT; U)
Host: buzzard.cs.unc.edu:8080
Accept: image/gif, image/x-xbitmap, image/jpeg,
        image/pjpeg, image/png, */*
Accept-Encoding: gzip
Accept-Language: en
Accept-Charset: iso-8859-1,*,utf-8
Cookie: SITESERVER=ID=8a064b785a043146e4599174a3d970
```

Response
headers

```
Date: Fri, 02 Feb 2001 19:10:11 GMT
Server: Apache/1.3.9 (Unix) (Red Hat/Linux)
Last-Modified: Tue, 30 Jan 2001 21:48:14 GMT
ETag: "1807135e-67-3a77369e"
Accept-Ranges: bytes
Content-Length: 103
Connection: close
Content-Type: text/plain
```